

Markscheme

Mathematics: analysis and approaches

Practice question bank

85. (a)

\$10 000 is invested under one of two options. Option A pays 5% p.a. simple interest. Option B pays 4.5% p.a. compounded annually.

Find the number of complete years after which Option B first exceeds Option A in value.

Option A: $A(n) = 10\,000(1 + 0.05n)$; Option B: $B(n) = 10\,000(1.045)^n$ (M1)

set up the inequality $B(n) > A(n)$ (M1)

check $n = 5$: $A = \$12\,500$, $B \approx \$12\,447$ (B still behind); check $n = 6$: $A = \$13\,000$, $B \approx \$13\,011$ (M1) **A1**

Option B first exceeds Option A after $n = 6$ complete years **A1**

[6 marks]